

Local gap statistics, telescoping, and normality

A local-pattern approach to Erdős problem 251

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Abstract

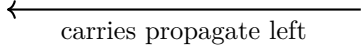
Erdős asked whether $\sum_{n \geq 1} p_n 2^{-n}$ is irrational. Under Kuperberg’s uniform Hardy–Littlewood conjecture, we prove that for each fixed integer $B \geq 2$ the corresponding series $\sum_{n \geq 1} p_n B^{-n}$ is normal to base B . Our local-pattern criterion classifies geometrically weighted series of periodic rational polynomials in consecutive gaps: each is normal or telescopes to a rational boundary value. The normal form determines all rational relations and joint orbit equidistribution of independent classes. For primes, a one-sided averaged hypothesis on tuples of size $O(\log \log X)$ suffices in each fixed degree and base range. The same criterion gives the classification unconditionally for rough integers in a subpower range. Already the degree-one input yields an uncountable rationally independent family of rounded gap-power series with jointly equidistributed orbits for every fixed finite subfamily. Telescopes are exactly the polynomials invisible to every one-point variation. One varying model point and a positive sieve bound produce an absolutely continuous measure dominating the phase limits. The exact recurrence supplies invariance; classical uniqueness then identifies the limits as Lebesgue measure. A finite version also gives the orbit star-discrepancy bound $D_N^* \ll_B (\log \log N)^{-1/2}$ for the linear prime series under Kuperberg’s conjecture.

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place value	2	1	2^{-1}	2^{-2}	2^{-3}	2^{-4}
$p_1/2$		1	0			
$p_2/2^2$			1	1		
$p_3/2^3$			1	0	1	
$p_4/2^4$				1	1	1
column sum	0	1	2	2	2	1
S_4 after carries	1	0	1	1	0	1



 carries propagate left

Figure 1: Binary addition of the first four terms: $S_4 = \sum_{n=1}^4 p_n 2^{-n} = (10.1101)_2$. Overlapping blocks produce column sums greater than one, so carries propagate left. The omitted infinite tail can change earlier digits.

1 Introduction and results

Erdős asked whether $\sum_{n \geq 1} p_n 2^{-n}$ is irrational, and also posed the question for $\sum p_n^d 2^{-n}$ [1, p. 103]. Here p_n is the n th prime. Under Kuperberg's uniform Hardy–Littlewood conjecture, we prove that $\sum p_n B^{-n}$ is normal to every fixed integer base $B \geq 2$: every finite digit word occurs with its expected frequency. This concerns a different number for each base. The coefficients are not digits: carries couple arbitrarily distant terms (Figure 1). We ask how much local arithmetic information determines the global digit distribution, and whether telescoping accounts for all polynomial exceptions. Summation by parts replaces positions by gaps $g_n = p_{n+1} - p_n$:

$$\sum_{n \geq 1} g_n B^{-n} = (B-1) \sum_{n \geq 1} p_n B^{-n} - p_1. \quad (1)$$

Tao suggested controlling about $\log \log n$ consecutive gaps by a sufficiently uniform prime-tuples hypothesis [2].

Nonconstant numerators need not give normal series: for every integer $B \geq 2$, telescoping gives unconditionally

$$\sum_{n \geq 1} (B g_n - g_{n+1}) B^{-n} = g_1 = 1.$$

The one-point test detects exactly these telescopes (Lemma 2.2). Algebra computes the local boundary relations; the arithmetic theorem excludes further rational relations in the stated range.

The basic recurrence. For the linear gap series,

$$U(n) = \sum_{i \geq 1} B^{-i} g_{n+i-1}, \quad U(n+1) = B U(n) - g_n.$$

Modulo one, the index shift multiplies by B . Rationality means eventual periodicity of this orbit; normality means equidistribution.

1.1 Main results

Throughout, unqualified logarithms are natural, $B \geq 2$ is an integer, and $e(t) = \exp(2\pi i t)$. Normality to base B is equivalent to equidistribution of $(B^N \alpha)_{N \geq 0}$ modulo one, by testing base- B cylinder intervals. Its Fourier form is Weyl's criterion [27]. For an increasing positive integer sequence a_n put $g_n = a_{n+1} - a_n$. Given a fixed period k , write $\bar{n} \in \{1, \dots, k\}$ for the

Numerator F	Normal form $\mathcal{N}_{B,1}F$	Series value or property
u_0	u_0	normal
u_0^2	u_0^2	normal
$u_0 u_1$	$u_0 u_1$	normal
$Bu_0 - u_1$	0	$g_1 = 1$
$u_0 + Bu_0^M - u_1^M$	u_0	$\alpha_{u_0} + 1$, normal

Table 1: Prime-gap examples at period one. Normality requires only $\deg(\mathcal{N}_{B,1}F)/\log B \leq \kappa$. The last row, for any fixed $M \geq 2$, therefore needs only the degree-one input. The rational boundary identities are unconditional.

representative of $n \bmod k$. For $F = (F_1, \dots, F_k)$, each $F_r \in \mathbb{Q}[u_0, \dots, u_w]$, define, whenever the series converge,

$$\alpha_F(a; B) = \sum_{n \geq 1} B^{-n} F_{\bar{n}}(g_n, \dots, g_{n+w}), \quad W_{r,d}(a; B) = \sum_{m \geq 0} g_{km+r}^d B^{-km-r}.$$

Reindexing exchanges a gap shift for a factor B and cycles the period label. The cyclic normal form $\mathcal{N}_{B,k}$ sends constant tuples to zero. If a nonconstant monomial lies in component r and its least gap index is j , shift that index to zero, move to component $\overline{r+j}$, and multiply by B^j ; extend linearly. Section 2 proves that this removes precisely the cyclic weighted telescopes.

The one-sided averaged Hardy–Littlewood hypothesis (AHL_κ), stated in (16), concerns tuples of size $O_\kappa(\log \log X)$ in windows of length $O_\kappa(\log X \log \log X)$. It asks only that the weighted sum of the adverse errors is $o(X/\log X)$. Their direction alternates with tuple order, including undercounts. No gap law or digital cancellation is assumed; κ fixes the available window depth. It implies the weaker positive-comparison condition (19) with $c = 1$, which suffices for the theorem below.

Theorem 1.1 (Periodic local prime-gap series). *Fix $\kappa > 0$ and assume (19) for the prime profile of Section 5 with this κ , for some fixed $c \geq 1$. Fix integers $B \geq 2$, $D, k \geq 1$ with $D/\log B \leq \kappa$. For every fixed rational local polynomial tuple F with $\deg \mathcal{N}_{B,k}F \leq D$, the actual prime-gap series α_F is rational if $\mathcal{N}_{B,k}F = 0$ and normal to base B^k (and hence to base B) otherwise. For any finite family in this range,*

$$\sum_i q_i \alpha_{F_i} \in \mathbb{Q} \iff \sum_i q_i \mathcal{N}_{B,k} F_i = 0 \quad (q_i \in \mathbb{Q}).$$

Independent normal forms give jointly equidistributed B^{kN} -orbits and rational linear independence with 1. Families with different fixed periods are first repeated into the least common multiple of their periods, where independence is tested.

Every separately fixed width, period and Fourier mode uses the same window parameters. For $w = 0$, the zero class means that every F_r is constant. Thus the theorem includes all nonconstant periodic single-gap polynomials and the joint orbit equidistribution of $W_{r,d}$. Kuperberg’s Conjecture 1.3 implies every fixed (AHL_κ) by Section 5, so all separately fixed degrees and bases are covered.

Corollary 1.2 (Erdős problem 251 and periodic position weights). *Under the arithmetic hypothesis of Theorem 1.1, for every integer $B \geq 2$ with $1/\log B \leq \kappa$ and every nonzero rational periodic sequence c_n , the series $\sum_{n \geq 1} c_n p_n B^{-n}$ is normal to base B . In particular, $\kappa \geq 1/\log 2$ implies the normality, and hence irrationality, of $\sum p_n 2^{-n}$. Kuperberg’s conjecture suffices.*

Section 3.3 proves this by finite Abel transformation. The square-gap series $\sum g_n^2/2^n$ is also covered by Theorem 1.1 in its degree-two range, unlike $\sum p_n^2/2^n$. For the linear prime series, Appendix B gives the quantitative bound (32) under Kuperberg's conjecture, using a finite version of the same positive-comparison input.

Theorem 1.3 (Unconditional rough-integer series). *There is an absolute finite A_* such that the classification, rational-relation and joint-orbit conclusions of Theorem 1.1 hold unconditionally for the gaps of the increasing enumeration of $\mathcal{R}_z = \{a : P^-(a) > z(a)\}$ if, for the fixed B, D ,*

$$\begin{aligned} z(x) &= \exp(\Psi(\log x)), & \Psi &\in C^1, & \Psi(t) &\rightarrow \infty, \\ 0 \leq t\Psi'(t) &\leq C_\Psi\Psi(t), & \frac{t}{\Psi(t)} &\geq A_* \frac{D}{\log B} \log(3\Psi(t)) & \text{eventually.} \end{aligned} \quad (2)$$

The constant is independent of separately fixed widths, coefficients and periods. Finite initial conventions are harmless.

Here $P^-(a)$ denotes the least prime factor for $a > 1$, and $P^-(1) := \infty$.

This includes $z(x) = x^{1/(C \log \log x)}$ for sufficiently large $C = C(B, D)$. If $\Psi(t) \log(3\Psi(t)) = o(t)$, the same rough sequence supports all separately fixed degrees and bases; for example, $z(x) = x^{1/(\log \log x \log \log \log x)}$ eventually. Different bases give different numbers, not absolute normality of one number. No fixed exponent $z(x) = x^\delta$, $\delta > 0$, is obtained.

1.2 Overview of the proof

Proposition 3.3 transfers local pattern and tail estimates to normality. Primes and rough integers are two instances. Integrality gives the recurrence; local variation gives regularity; the arithmetic comparison transports it; invariance identifies Lebesgue measure. The polynomial algebra decides when the local variation is present.

Local variation. In the sieve model, not the prime sequence, fix all survivors except x_j . The remaining configuration is a complete frame. The adjacent gaps $v, W - v$ contribute

$$B^{-j-1}(Bv + W - v) = B^{-j-1}((B-1)v + W).$$

This nonzero slope is the linear source of regularity (Figure 2).

Regularity and invariance. Averaging complete frames without a slot normalization, the Selberg bound in Section 4 dominates positive phase tests by interval integrals. The resulting dominating measure is absolutely continuous, excluding a rational value with its finite orbit. The recurrence supplies invariance; the classical principle behind hot-spot criteria (Lemma 3.2) then gives normality. No prime exponential-sum estimate is used as arithmetic input.

The exact exceptions. Lemma 2.2 identifies invisibility with telescoping to a rational boundary. Otherwise the leading action has an absolutely continuous interval image on almost every limiting frame. The first-gap mean keeps the dominating measure finite; the k -step recurrence handles periodic coefficients.

Arithmetic and scale. The normal-form degree d requires rank $j = d \log_B G + O(1)$; telescoping terms cost no depth. For primes, $G = \log X$ explains the $\log \log X$ depth and $D/\log B \leq \kappa$; the reserve controls the infinite tail. Stopped Bonferroni converts the one-sided tuple hypothesis to pattern comparison; first-gap averages bound the tail (Section 5). Appendix A supplies these inputs unconditionally.

1.3 Related work

Land independently proved conditional irrationality of $\sum p_n 2^{-n}$ under Kuperberg's conjecture before the present work appeared [4]. He selects rare prime patterns and averages to control

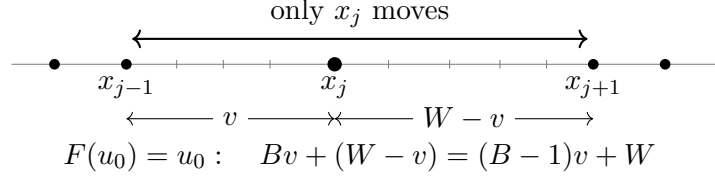


Figure 2: One-point variation in the sieve model. Ticks mark eligible presieve sites between the fixed neighbours. The proof averages unnormalized insertion sums over all complete frames, including empty slots (see (12)). The linear action has nonzero slope for $B > 1$.

the tail; our local averaging yields normality and the polynomial classification. No implication between the two specialized local hypotheses is claimed.

Gallagher obtains Poisson statistics from uniform fixed-order tuple estimates [5]; our tests need growing order, as allowed by Kuperberg's conjecture [7]. Tao also compares primes with a random sieve [8]. Banks–Ford–Tao's signed averaged input [9] does not automatically control our one-sided error norm.

Copeland–Erdős concatenate primes without carries [10]; Nakai–Shiokawa extend this to positive integer-valued nonconstant polynomials evaluated at primes [11], and Madritsch treats rounded pseudo-polynomials, including nonintegral powers [12]. These are concatenations, whereas our gap series have overlapping contributions and carries. Bailey–Crandall and Lagarias connect arithmetical series with digit orbits [13, 14]; we instead establish regularity by positive comparison, without verifying their Hypothesis A. Bailey–Crandall also construct uncountable normal families [13, Theorem 4.12]. Corollary 3.4 provides a concrete gap-parametrized family with rational linear independence and joint orbit equidistribution of every fixed finite subfamily.

Kováč's variable-denominator counterexample chooses integers $q_n \geq 2$, $q_n = o(p_n)$, without requiring monotonicity, so that $\sum p_n/(q_1 \cdots q_n)$ telescopes [3]. Our geometric weights, in contrast, are fixed in advance.

Pratt proved irrationality of $\sum \omega(n)2^{-n}$ conditionally on uniform prime tuples [17]; Tao–Teräväinen proved it unconditionally [18]. Here $\omega(n)$ counts distinct prime factors. Their multiplicative correlation input does not supply our rank-dependent prime-pattern comparison. Figure 4 summarizes the connections behind the proof.

2 Cyclic telescopes and the one-point test

Let $\mathcal{G} = \mathbb{Q}[u_0, u_1, \dots]$ with finite variable support and $Su_i = u_{i+1}$. The combined shift on $\mathcal{G}^k$ is $(AH)_r = SH_{r+1}$, and

$$\mathcal{T}_B = B - A, \quad (\mathcal{T}_B H)_r = BH_r - SH_{r+1}.$$

For example, at period $k = 2$, $A(u_0, 0) = (0, u_1)$. The local cohomological equation $(B - A)H = F$ asks for a polynomial primitive with finite variable support. The same geometric reindexing gives the finite boundary formula (3) and the k -step recurrence (6). The normal form below computes the obstruction to local solvability; the converse from rationality of the series will require arithmetic.

Write $e_r M$ for the tuple with M in component r and zero elsewhere. For a nonconstant monomial $M = S^j M^b$ with $u_0 \mid M^b$, define

$$\mathcal{N}_{B,k}(e_r M) = B^j e_{r+j} M^b, \quad \mathcal{N}_{B,k}(\text{constant tuple}) = 0.$$

Degree of a tuple means the maximum component degree, with $\deg 0 = -\infty$.

Lemma 2.1 (Cyclic normal form). *Every $F \in \mathcal{G}^k$ has a unique decomposition $F = \mathcal{N}_{B,k}F + \mathcal{T}_B H$. It preserves degree bounds and $\ker \mathcal{N}_{B,k} = \mathcal{T}_B \mathcal{G}^k$.*

Proof. Every nonconstant labelled monomial belongs to exactly one chain $\mathbf{A}^j C$ ($j \geq 0$), starting at $C = e_q M^b$ with $u_0 \mid M^b$: indeed $\mathbf{A}^j C = e_{q-j} \mathbf{S}^j M^b$. Thus uniquely $F = c + \sum_C p_C(\mathbf{A})C$, with a constant tuple c . Ordinary polynomial division gives

$$p_C(z) = p_C(B) + (B - z)q_C(z), \quad \mathcal{N}_{B,k}F = \sum_C p_C(B)C.$$

The primitive is $\sum_C q_C(\mathbf{A})C + H_c$, where

$$H_c = (B^k - 1)^{-1} \sum_{i=0}^{k-1} B^{k-1-i} \mathbf{A}^i c.$$

Here $\mathbf{A}^k c = c$. Division on each chain and this inverse on constants prove uniqueness. Gap degree is constant along a chain, so both operations preserve degree bounds. If F uses only indices at most w , every nonconstant term of H uses indices at most $w - 1$. Moreover, $\mathcal{N}_{B,k} \mathbf{A} = B \mathcal{N}_{B,k}$ and the kernel is $\mathcal{T}_B \mathcal{G}^k$. $\square$

Since $\mathcal{N}_{B,k}$ cannot increase degree and is unchanged by subtracting $\mathcal{T}_B H$, the canonical decomposition attains the least degree in each nonzero class modulo telescopes.

Evaluation gives the finite identity

$$\alpha_F^{[N]} - \alpha_{\mathcal{N}_{B,k}F}^{[N]} = H_1(g_1, \dots) - B^{-N} H_{N+1}(g_{N+1}, \dots), \quad (3)$$

where superscript $[N]$ denotes truncation at $n = N$. Whenever the endpoint vanishes, the series difference is rational; Lemma 3.1 ensures this under (S). Thus the series modulo $\mathbb{Q}$ depends only on the vector-space class of F modulo $\mathcal{T}_B \mathcal{G}^k$. Within the effective-degree range, the arithmetic theorem rules out further rational relations and proves normality for every nonzero class. Without arithmetic the converse fails: $a_n = n^2$ gives $\alpha_{u_0} = (3B - 1)/(B - 1)^2 \in \mathbb{Q}$ although $\mathcal{N}_{B,1} u_0 = u_0$.

In the action below, shifts also act on integer-indexed gap variables.

Lemma 2.2 (Telescopes and local invisibility). *For $F \in \mathcal{G}^k$ using $u_0, \dots, u_w$, set $u_{-1} = v$, $u_0 = W - v$ and define*

$$\pi_s(v; y) = \sum_{t=-w-1}^0 B^{-t} F_{s+t+1}(u_t, \dots, u_{t+w}), \quad (4)$$

where y consists of W and the exterior gaps. Then

$$\mathcal{N}_{B,k}F = 0 \iff F = \mathcal{T}_B H \text{ for some } H \iff \partial_v \pi_s \equiv 0 \text{ for every } s,$$

as polynomial identities in v, y . Moreover, if $F = \mathcal{N}_{B,k}F \neq 0$ has degree d , some action has a degree- d homogeneous part nonconstant in v .

Proof. First let $F = \mathcal{N}_{B,k}F \neq 0$ have degree d , and let $P = F^{[d]}$ be its top homogeneous part. The normal form acts degree by degree, so P is again normal. Write $\pi_s(P)$ for its action: the linear substitution makes this exactly the degree- d part of $\pi_s(F)$. Let h be the greatest variable index of P . If $h \geq 1$, choose a component r with $\partial_{u_h} P_r \neq 0$ and put $s = r + h$. Before substituting $u_{-1} = v$, $u_0 = W - v$, the derivative $(\partial_{u_{-1}} - \partial_{u_0})\pi_s(P)$ has just one term depending on u_{-h-1} :

$$B^{h+1} \mathbf{S}^{-h-1} \partial_{u_h} P_r.$$

It really depends on that variable, since every normal monomial contains u_0 . Earlier summands $t < -h - 1$ use only $u_t, \dots, u_{t+h}$, so neither moved gap occurs and their derivatives vanish. Thus the derivative is nonzero, also after the invertible substitution. If $h = 0$, all derivatives could vanish only if $BP'_s(u_{-1}) = P'_{s+1}(u_0)$ for every s . The independent variables force constant derivatives a_s with $Ba_s = a_{s+1}$; hence $(B^k - 1)a_s = 0$, contradicting $P \neq 0$.

For general F , use the normal-form decomposition. A telescope $F = \mathcal{T}_B H$ contributes only the exterior boundary

$$\pi_s(F) = B^{w+2}(\mathcal{S}^{-w-1}H)_{s-w} - (SH)_{s+2};$$

if H is not constant its largest index is at most $w - 1$. The normal-part argument and Lemma 2.1 prove both equivalences. $\square$

This is a weighted polynomial instance of the discrete variational principle that a density invisible to all interior variations is a boundary term; compare [26]. The direct proof above requires none of the general variational machinery.

3 A common digital criterion

3.1 Pattern assumptions and rigidity

Let a_n increase through positive integers and fix a real auxiliary scale $\rho > 1$, not necessarily an integer digit base; write $\kappa = 1/\log \rho$. For every sufficiently large integer X set

$$\begin{aligned} G_X &\rightarrow \infty, & L = L_X &= \left\lceil \log_\rho G_X + \sqrt{\log_\rho G_X} \right\rceil, & S = S_X &= \lfloor (6/5)LG_X \rfloor, \\ I_X &= \{n : X < a_n \leq 2X\}, & m_X &= |I_X| > 0, & T_\rho(n) &= \sum_{h \geq 1} g_{n+h-1} \rho^{-h}. \end{aligned} \quad (5)$$

Here n is a sequence index and X a value bound; L counts gaps, whereas S is a physical span. For a nonempty finite index set I , write $\text{Avg}_{n \in I} f(n) = |I|^{-1} \sum_{n \in I} f(n)$. Write $V(y) = \prod_{p \leq y} (1 - 1/p)$. The rooted sieve law $\mathbb{P}_{0,y}$ excludes one independent uniform nonzero residue at each prime $p \leq y$; write its nonnegative survivors as $0 = x_0 < x_1 < \dots$. Thus the origin, or root, is always retained.

(S) compares initial patterns with a calibrated sieve; (T) bounds the tail in mean. Neither assumption mentions digits. There are probability mixtures ω_X of these laws with $\sup_{y \in \text{supp } \omega_X} |V(y)^{-1}/G_X - 1| \rightarrow 0$ such that, uniformly over complex first- L tests f with $|f| \leq 1$ and $f = 0$ when $x_L > S$,

$$\text{Avg}_{n \in I_X} f(a_{n+1} - a_n, \dots, a_{n+L} - a_n) = \int \mathbb{E}_{0,y} f(x_1, \dots, x_L) d\omega_X(y) + o(1), \quad (S)$$

$$\text{Avg}_{n \in I_X} T_\rho(n + L) = O_\rho(G_X). \quad (T)$$

The test may depend on X . The infinite remainder (T) is about the actual sequence, not an infinite model observable. Its series is well-defined by the following consequence of (S); the quantitative mean bound in (T) remains a separate assumption. The model estimates used here are proved in Section 4.

Lemma 3.1 (Growth from local pattern comparison). *Condition (S) implies $m_X \rightarrow \infty$ and $\log a_n/n \rightarrow 0$. Consequently every fixed local polynomial series with geometric denominator converges absolutely, and all boundaries in (3) vanish.*

Proof. (S) and the model span bound make the actual mass on $x_L \leq S$ tend to one. A fixed location $1 \leq d \leq S$ survives the independent prime coordinates $S < p \leq y$ with probability

$$\prod_{S < p \leq y} (1 - 1/(p-1)) \leq V(y)/V(S) \ll \log S/G_X = o(1),$$

uniformly in calibrated mixtures. Choose $n \in I_X$ with $a_{n+L} - a_n \leq S$, and put $d_X = g_n$. (S) on $\mathbf{1}_{\{x_1=d_X, x_L \leq S\}}$ gives $1/m_X \leq \epsilon_X + O(\log S/G_X) \rightarrow 0$, where $\epsilon_X \rightarrow 0$ is the uniform error in (S). Thus $m_X \rightarrow \infty$, without using a spacing limit.

Fix K . For all sufficiently large integers J , $A(2^J) \geq K(J - J_0)$, by summing over disjoint dyadic intervals in the values of the sequence; here $A(x) = \#\{n : a_n \leq x\}$. If $2^J \leq a_n < 2^{J+1}$, then $\log a_n \leq (n/K + J_0 + 1) \log 2$. Let $n \rightarrow \infty$, then $K \rightarrow \infty$. Every fixed shifted polynomial value has absolute value at most $\exp(o(n))$, which proves the last assertions. No mean-tail estimate follows from this growth argument alone. $\square$

Write λ for normalized Lebesgue measure on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $T_C x = Cx \bmod 1$. The following classical principle also underlies hot-spot criteria [15, Lemma 3.2].

Lemma 3.2 (Absolute continuity, invariance and rational lifts). *For an integer $C \geq 2$, a T_C -invariant probability $\mu \ll \lambda$ equals λ . If $q\alpha$ is normal to base C for an integer $q \geq 1$, then α is normal to base C . Consequently rational affine maps with nonzero slope preserve normality.*

Proof. For $t \in \mathbb{Z} \setminus \{0\}$, invariance and Riemann–Lebesgue give $\hat{\mu}(t) = \hat{\mu}(C^h t) \rightarrow 0$, so $\mu = \lambda$. If $q\alpha$ is normal, any empirical limit ν of the C -orbit of α is invariant and $(T_q)_* \nu = \lambda$. For compact λ -null K , its image $T_q K$ is compact and null, and

$$\nu(K) \leq \nu(T_q^{-1}(T_q K)) = \lambda(T_q K) = 0.$$

Inner regularity gives $\nu \ll \lambda$, hence $\nu = \lambda$. Integer multiples preserve normality by Weyl's criterion. For a rational affine map, clear denominators and use this lift. $\square$

The last assertion is Wall's normality-preservation theorem [25]; see also [28, Proposition 4.25 and Remark 4.28]. Lyons constructed a measure with vanishing Fourier coefficients carried by numbers not normal to base 2 [16]. Here the additional invariance identifies the limiting measure itself with Lebesgue measure.

3.2 The digital transfer criterion

Proposition 3.3 (Digital transfer criterion). *Assume $(S)/(T)$. Fix integers $B \geq 2$, $D, k \geq 1$ with $B \geq \rho^D$. For any rational local tuple $F \in \mathcal{G}^k$ with $\deg \mathcal{N}_{B,k} F \leq D$, its series α_F is rational if $\mathcal{N}_{B,k} F = 0$, and normal to base B^k (and hence to base B) otherwise.*

Proof. Lemma 3.1 justifies all polynomial series and boundaries. Apply Lemma 2.1, (3) and the rational-affine lift in Lemma 3.2; it suffices to treat an integral normal tuple $F = \mathcal{N}_{B,k} F \neq 0$ of degree $d \leq D$ and fixed width $w + 1$.

Step 1. Recurrence and truncation. Set

$$U(n) = \sum_{i \geq 1} B^{-i} F_i(g_{n+i-1}, \dots, g_{n+i+w-1}).$$

Then $U(1) = \alpha_F$, and exact reindexing gives

$$U(n+k) = B^k U(n) - \sum_{i=1}^k B^{k-i} F_i(g_{n+i-1}, \dots, g_{n+i+w-1}), \quad (6)$$

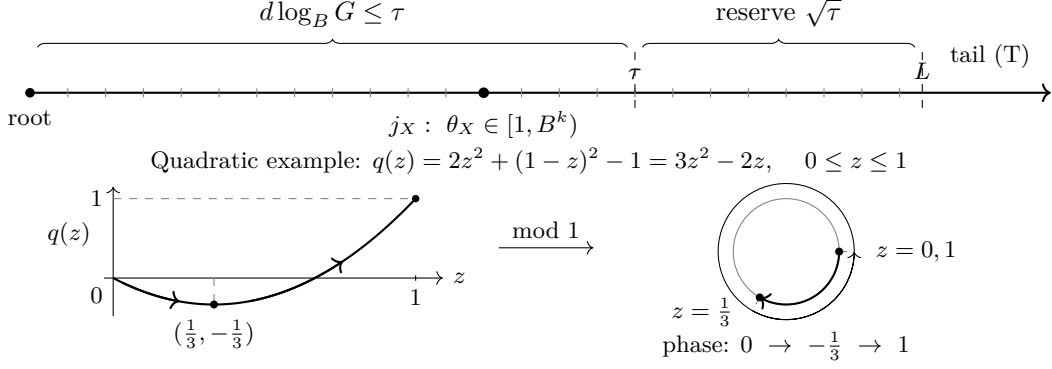


Figure 3: Scale selection and a quadratic local action. Above, $\tau = \log_\rho G$ and $L = \lceil \tau + \sqrt{\tau} \rceil$; the reserve accommodates fixed width and period and ensures $G\rho^{-L} \rightarrow 0$. Below, $F(u_0) = u_0^2$, $B = 2$, unit normalized insertion span and $\theta_X = 1$ give q up to an additive constant. The phase reverses at $z = 1/3$; directed traces on the circle are offset for visibility. The image of uniform length is absolutely continuous, but not uniform: invariance is still needed.

where the subtracted sum is an integer. The labels are relative to i , not n .

For the window length $L = L_X$, put $K = L - w$ and write the truncation as a function of the first L offsets, with $h_i = x_{i+1} - x_i$:

$$\Phi_K(x) = \sum_{i=1}^K B^{-i} F_i(h_{i-1}, \dots, h_{i+w-1}).$$

At the actual root n substitute $x_i = a_{n+i} - a_n$ and write $\Phi_K(n)$.

Write $G = G_X$. For positive integer gaps, $|F_r(z)|^{1/d} \ll_F \sum_j z_j$. Subadditivity, $B^{-i/d} \leq \rho^{-i}$ and reindexing give

$$|U(n) - \Phi_K(n)|^{1/d} \leq c_{F,\rho} \rho^{-K} T_\rho(n + K).$$

Put $\psi(t) = \min(1, t)$. Shifting the bounded sequence $\psi(c_{F,\rho} \rho^{-K} T_\rho(n + K))$ by $w = L - K$ costs at most $2w/m_X$, so (T) and $\rho^{-K} = \rho^w \rho^{-L}$ give

$$\text{Avg}_{n \in I_X} \psi(|U(n) - \Phi_K(n)|^{1/d}) \leq c_{F,\rho} \rho^{-K} \text{Avg}_{n \in I_X} T_\rho(n + L) + \frac{2w}{m_X} \rightarrow 0.$$

Here $G\rho^{-L} \leq \rho^{-\sqrt{\log_\rho G}} \rightarrow 0$. Uniform continuity transfers every fixed continuous phase test. Condition (S) bounds the mass omitted by the span restriction.

Step 2. One-point variation at the critical rank. Moving a point x_j , with $w + 1 \leq j \leq K - 1$, changes the phase by its variable part $B^{-j-1} \pi_s(v; y)$, where $s \equiv j \pmod{k}$, $v = x_j - x_{j-1}$, and y consists of $W = x_{j+1} - x_{j-1}$ and the interacting exterior gaps.

Apply Lemma 2.2 to the normal tuple. For some residue s , the degree- d homogeneous part of π_s is nonconstant in v . This single check covers all periods and all fixed local widths.

Put $C = B^k$. Choose once $j_X \equiv s \pmod{k}$ with

$$\theta_X = G^d B^{-j_X-1} \in [1, C).$$

The degree/base inequality and the reserve give $w + 1 \leq j_X \leq L - w - 1$ eventually. This rank is independent of the Fourier mode and accuracy (Figure 3).

In the model, let $\mathcal{A}$ be the sites in $[1, S]$ surviving primes up to S , and N the final survivor count. The auxiliary law $\mathbb{E}^-$ retains the original $(y, \mathcal{A}, N)$ -marginal, restricts to $N \geq L$, and chooses a uniform $(N - 1)$ -subset U^- of $\mathcal{A}$, without normalizing the restricted mass. It differs

from fixed-rank deletion; (12) accounts for the change of law. The auxiliary local frame Y_X^- consists of the gaps of U^- at ranks $j_X - w - 1, \dots, j_X + w - 1$, divided by G . These ranks lie in $0, \dots, L - 2$, so Lemma 4.2 gives tightness and absolutely continuous limits. Write $W(Y)$ for its central gap (the insertion span) and $Q_Y(z)$ for the top homogeneous action in $z = v/G$. Write $\Phi_K(\mathcal{F}, u)$ for evaluation after ordered insertion of u at rank j_X ; retain repeated entries at slot endpoints. Define the outer phase on every frame, even one with no eligible insertion, by polynomial evaluation at its left endpoint $a_{\mathcal{F}}$:

$$O_{\mathcal{F}} = \Phi_K(\mathcal{F}, a_{\mathcal{F}}) - B^{-j_X-1} \pi_s(0; y_{\mathcal{F}}) \pmod{1}.$$

This permits the virtual zero gap. On compact frames, uniformly for $z = (u - a_{\mathcal{F}})/G \in [0, W(Y)]$,

$$\Phi_K(\mathcal{F}, u) = O_{\mathcal{F}} + \theta_X Q_Y(z) + o(1) \pmod{1}.$$

The outer phase may vary arbitrarily between frames, but is independent of the inserted point on each frame.

Step 3. Positive domination and absolute continuity. Put $J_X = \{n \in I_X : n \equiv 1 \pmod{k}\}$ and $\ell_X = |J_X| = m_X/k + O(1) \rightarrow \infty$. Consider any weak limit ν of the class measures

$$\nu_X = \ell_X^{-1} \sum_{n \in J_X} \delta_{U(n) \bmod 1}.$$

Along a further subsequence the joint auxiliary law of $(O_{\mathcal{F}}, Y_X^-, \theta_X)$ converges to a probability Λ : the circle and $[1, C]$ are compact, and the frame laws are tight. Place the vanishing missing mass at a fixed point. This extraction precedes all tests and cutoffs. The one-point lemma gives a nonzero polynomial in Y among the coefficients of $\partial_z Q_Y$. Its zero set is Lebesgue-null, so the absolutely continuous Y -marginal makes Q_Y nonconstant Λ -almost surely. Since $W \geq 0$ is continuous, the gap means and Portmanteau give $\int W(Y) d\Lambda \leq 2$.

Define a finite measure σ on $\mathbb{T}$ by

$$\sigma(f) = \int \int_0^{W(Y)} f(O + tQ_Y(z)) dz d\Lambda(O, Y, t).$$

Then $\sigma(\mathbb{T}) \leq 2$ and $\sigma \ll \lambda$. A nonconstant polynomial has finitely many critical points; change of variables on its monotone pieces proves absolute continuity of its interval image. Translation and reduction modulo one preserve it.

Fix $f \geq 0$ continuous on $\mathbb{T}$. Positivity gives $\nu_X(f) \leq (k + o(1)) \text{Avg}_{n \in I_X} f(U(n))$. Choose a continuous compactly supported $0 \leq \chi_A \leq 1$ equal to one on $[0, A]^{2w+1}$. In an original model configuration, delete x_{j_X} and denote the resulting local frame coordinates, divided by G , by Y^{del} . The central gap is a sum of two original gaps; the others are single gaps. The original rooted means therefore give

$$\mathbb{E} \left[\mathbf{1}_{\{N \geq L\}} \sum_i Y_i^{\text{del}} \right] \leq (2w + 2)(1 + o(1)).$$

Markov bounds the discarded original frame mass by $O_w(A^{-1})$. Since $Y^{\text{del}}(\mathcal{F} \cup \{u\}) = Y(\mathcal{F})$ for every admissible insertion, the same cutoff becomes $\chi_A(Y_X^-)$ under the auxiliary law. Thus

$$\nu_X(f) \leq 24k \mathbb{E} \left[\chi_A(Y_X^-) \int_0^{W(Y_X^-)} f(O_{\mathcal{F}} + \theta_X Q_{Y_X^-}(z)) dz \right] + O_{F,k}(A^{-1}) \|f\|_{\infty} + o(1).$$

Here (S), the span estimate and the remainder bound first transfer the test on actual offsets to the model. The probabilities of exceptional counts and large original frames are bounded

before changing measure. On the compact cutoff the tests are equicontinuous, uniformly over the outer phase. Equations (12) and (13) give $12\partial G/\log G < 24$; positivity permits extension to all auxiliary frames. The cutoff integral is bounded and jointly continuous in (O, Y, t) . Pass to the fixed joint limit, use $\chi_A \leq 1$, and then send $A \rightarrow \infty$. This gives $\nu \leq 24k\sigma$, hence $\nu \ll \lambda$. *Step 4. Invariance and the full orbit.* For every continuous f , (6) gives

$$|\nu_X(f(C \cdot)) - \nu_X(f)| \leq 2\|f\|_\infty/\ell_X \rightarrow 0.$$

Thus ν is T_C -invariant. Lemma 3.2 gives $\nu = \lambda$, and consequently $\nu_X \Rightarrow \lambda$.

Fix $q \in \mathbb{Z} \setminus \{0\}$ and $\epsilon > 0$, and choose X_0 so that every integer $X \geq X_0$ has $|\text{Avg}_{n \in J_X} e(qU(n))| \leq \epsilon$. Starting at $b = a_{1+k(N-1)}$, repeatedly take $X = \lfloor b/2 \rfloor$, remove the terms with values in $(X, 2X]$, and replace b by X ; stop when $b < 2X_0$. Each step omits at most one integer. Lemma 3.1 bounds the number of steps by $O(\log a_{1+k(N-1)}) = o(N)$. The complete blocks contribute at most ϵN , and the final bounded interval contributes $O_{X_0}(1)$. Hence

$$\frac{1}{N} \sum_{h=0}^{N-1} e(qU(1+kh)) \rightarrow 0.$$

By (6), these are the character means of the C -orbit of α_F . Weyl's criterion proves normality to $C = B^k$. For $0 \leq r < k$, multiplication by B^r preserves Lebesgue measure, so $(B^{kh+r}\alpha_F)_h$ is equidistributed. Interleaving these k subsequences gives base- B normality. $\square$

Positive comparison suffices. The criterion and its consequences also hold if (S) is replaced by its upper inequality for $0 \leq f \leq 1$, with the model term multiplied by a fixed $c \geq 1$, provided the actual mass outside $x_L \leq S$ is $o(1)$. Keep the same support, uniformity and calibrated mixtures, and (T). The positive point test still proves Lemma 3.1; Step 3 gives $\nu \leq 24kc\sigma$. All other steps are unchanged.

3.3 Relations, dimension and positions

The rational boundary in (3) and the criterion give exactly the rational relations between the normal classes. Under (S)/(T), fix integers $B \geq 2$, $k, H, D \geq 1$ with $B \geq \rho^D$. For period k , H gap variables and total degree at most D , the normal monomials are the k labelled copies of $u_0^{\beta_0} \cdots u_{H-1}^{\beta_{H-1}}$ with $\beta_0 \geq 1$ and $\sum \beta_i \leq D$. Removing one factor u_0 gives

$$\dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \{1, \alpha_F : F_r \in \mathbb{Q}[u_0, \dots, u_{H-1}], \deg F_r \leq D\} = 1 + k \binom{H+D-1}{D-1}.$$

Every nonzero integer combination of independent normal classes is normal to B^k ; scalar Weyl proves the joint orbit assertion. At $H = 1$ these classes are precisely the kD components $W_{r,d}$.

For different fixed periods k_i , put $k^* = \text{lcm}(k_i)$ and repeat each tuple into its k^* components. This embedding commutes with $\mathbf{A}$ and the normal form: it maps normal tuples to normal tuples and telescopes to telescopes. It changes no series value. Test independence after this embedding and apply the same argument at period k^* along the powers B^{k^*N} .

Regard the rational periodic position weights c_n as a constant k -tuple. Using the inverse already computed in Lemma 2.1, set $d = (B - \mathbf{A})^{-1} \mathbf{A} c$, with periodic indices. On this constant space $\mathbf{A}^k = 1$, and both factors are invertible. Thus $c \mapsto d$ is invertible and $Bd_j - d_{j+1} = c_{j+1}$. Finite telescoping gives

$$\sum_{n=1}^N c_n a_n B^{-n} = a_1 d_0 + \sum_{j=1}^{N-1} d_j g_j B^{-j} - a_N d_N B^{-N}.$$

The endpoint vanishes by Lemma 3.1. Thus nonzero periodic position weightings are normal, and their residue components inherit common- B^k joint equidistribution by the scalar character test and rational-affine invariance. Equation (1) is the period-one case; no new position-tail estimate is needed.

3.4 Rounded real gap powers

Corollary 3.4 (Rounded real gap powers). *Assume (S)/(T), or the positive-comparison variant of Proposition 3.3. Fix integers $B \geq 2$, $k \geq 1$ with $B \geq \rho$. For $1 \leq r \leq k$ and real $0 < \alpha \leq 1$, put*

$$V_{r,\alpha} = \sum_{h \geq 0} \lfloor g_{kh+r}^\alpha \rfloor B^{-kh-r}.$$

Here α denotes an exponent, distinct from the series notation α_F . Every nontrivial finite rational linear combination of this indexed family is normal to base B^k . Every fixed finite subfamily has jointly equidistributed B^{kN} -orbits, and 1 together with the entire family is rationally linearly independent. The same assertions hold with all floors replaced by ceilings. Moreover, for a nonconstant $P \in \mathbb{Q}[T]$ of degree m and fixed $0 < \alpha \leq 1/m$, both $\sum_{n \geq 1} P(\lfloor g_n^\alpha \rfloor) B^{-n}$ and its ceiling analogue are normal to base B .

Proof. Clear denominators and group equal exponents, writing $F_r(q) = \sum_\ell b_{r\ell} \lfloor q^{\alpha_\ell} \rfloor$ with integer coefficients. Let $a > 0$ be the largest exponent with a nonzero coefficient column (b_r) . Lemma 3.1 and $|F_r(q)| \ll q$ for integers $q \geq 1$ give convergence. Define U, Φ_L by the sums in Step 1 of Proposition 3.3, with width zero, and put $G = G_X$. The recurrence (6) is exact with an integer subtracted sum, and

$$|U(n) - \Phi_L(n)| \ll \rho^{-L} T_\rho(n + L).$$

Thus (T) and $G\rho^{-L} \rightarrow 0$ transfer continuous phase tests using only the degree-one remainder estimate.

For each fixed A , uniformly over integers $0 \leq q \leq AG$, $G^{-a}F_r(q) = b_r(q/G)^a + o(1)$, with $F_r(0) = 0$. Only two summands change on moving a point. Choose a residue s as below and $j_X \equiv s \pmod{k}$ with $\theta_X = G^a B^{-j_X-1} \in [1, B^k)$. Since $0 < a \leq 1$ and $B \geq \rho$, the window reserve gives $1 \leq j_X \leq L-1$ eventually. The rounding contribution to the local phase is $O(B^{-j_X-1}) = O(G^{-a})$. Retaining the unchanged summands in $O_{\mathcal{F}}$, the phase on compact normalized frames is $\Phi_L(\mathcal{F}, u) = O_{\mathcal{F}} + \theta_X Q_W(z) + o(1) \pmod{1}$, where

$$Q_W(z) = Bb_s z^a + b_{s+1} (W - z)^a, \quad 0 \leq z \leq W.$$

Here W is the normalized insertion span and $z = v/G$. For $a = 1$, some s has $Bb_s - b_{s+1} \neq 0$ by cyclicity. For $a \neq 1$, choose $b_s \neq 0$. The equation $Q'_W(z) = 0$ has at most one solution in $0 < z < W$, since $(z/(W-z))^{a-1}$ is strictly monotone. Hence for every $W > 0$ the interval image is absolutely continuous. Endpoint singularities when $a < 1$ are harmless: the actions are uniformly continuous on compact normalized frames, and change of variables applies on their open monotonicity intervals.

Use the same cutoff and joint frame-outer-phase extraction as in Step 3. Lemma 4.2 gives $\int W d\Lambda \leq 2$; zero-length slots contribute no mass. Equations (12) and (13) therefore give $\nu \leq 24kc\sigma \ll \lambda$, with $c = 1$ under (S). No independence of frame and outer phase is needed. The recurrence, Lemma 3.2, and the full-orbit argument in Step 4 prove normality. Rational lifts and scalar Weyl give the remaining conclusions.

Ceilings have the same bounded rounding error. For the polynomial assertion, clear denominators in P , let $b \neq 0$ be its leading coefficient, and put $a = m\alpha \leq 1$. With R denoting

either floor or ceiling, $F(q) = P(R(q^\alpha)) - P(0)$ is integral, satisfies $|F(q)| \ll q$, and, uniformly for integers $0 \leq q \leq AG$,

$$G^{-a}F(q) = b(q/G)^a + O_A(G^{-\alpha}).$$

The preceding $k = 1$ argument applies; restore the rational constant $P(0)/(B - 1)$ and lift back through the cleared denominator. $\square$

The prime hypothesis in Theorem 1.1, and hence also (AHL_κ) , supplies these conclusions when $1/\log B \leq \kappa$; for the rough sequence they hold unconditionally under (2) with $D = 1$. Thus $\sum \lfloor \sqrt{g_n} \rfloor B^{-n}$ is normal in the same degree-one range. Each exponent and tested finite family is fixed before taking limits; no uniform rate in α is asserted.

4 The finite rooted sieve model

We prove the criterion's model estimates; Section 5 and Appendix A then verify (S)/(T) for primes and rough integers.

We use the classical PNT and Mertens estimates [22, 21]; for the latter see also [23, Theorem 2.7(d),(e)]:

$$p_n \sim n \log n, \quad V(y) \log y \longrightarrow e^{-\gamma} > 1/2, \quad \sum_{p \leq y} p^{-1} = \log \log y + O(1). \quad (7)$$

Here γ is Euler's constant; $\gamma \leq H_5 - \log 5 < \log 2$ proves the displayed numerical inequality. For a function of independent finite coordinates whose coordinate sensitivities are c_i , we use only

$$\text{Var}(Z) \leq \frac{1}{4} \sum_i c_i^2.$$

Its orthogonal Doob differences have conditional ranges at most c_i , hence conditional variances at most $c_i^2/4$. The other model input is the following elementary Selberg quadratic [23, Section 3.2].

Lemma 4.1 (Uniform finite Selberg upper bound). *If $\mathcal{A}$ lies in an interval of s consecutive integers and, for an integer $1 \leq R \leq s$, avoids one residue class modulo every prime $p \leq R$, then*

$$|\mathcal{A}| \leq s/J(R) + R^2, \quad J(R) = \sum_{1 \leq d \leq R} \frac{\mu(d)^2}{\phi(d)} \geq \log(R+1).$$

Taking $R = \lfloor \sqrt{s}/\log s \rfloor$ gives $|\mathcal{A}| \leq (2 + o(1))s/\log s$, uniformly in the classes and interval.

Proof. Here μ is the Möbius function and ϕ is Euler's totient. Choose by CRT an integer c in all forbidden classes and put

$$y_r = \frac{\mu(r)}{\phi(r)J(R)}, \quad \lambda_d = d \sum_{m \leq R/d} \mu(m)y_{dm}.$$

Finite divisor inversion gives $\lambda_1 = 1$ and $\sum_{r|d, d \leq R} \lambda_d/d = y_r$. Expand the majorant $(\sum_{d \leq R, d|n-c} \lambda_d)^2$ and count each residue class in the interval:

$$|\mathcal{A}| \leq s \sum_{d,e \leq R} \frac{\lambda_d \lambda_e}{[d,e]} + \left(\sum_{d \leq R} |\lambda_d| \right)^2.$$

The gcd identity $(d, e) = \sum_{r|d, r|e} \phi(r)$ makes the main quadratic $\sum_r \phi(r) y_r^2 = 1/J(R)$. For squarefree d ,

$$|\lambda_d| = \frac{d}{\phi(d)J(R)} \sum_{\substack{m \leq R/d \\ (m, d)=1}} \frac{\mu(m)^2}{\phi(m)} \leq 1 :$$

in $J(R)$ retain the distinct terms em , with $e | d$, $(m, d) = 1$, $m \leq R/d$, and use $\sum_{e|d} \phi(e)^{-1} = d/\phi(d)$. Nonsquarefree d have zero weight, so the error is at most R^2 . Finally, the reciprocal sum of all integers with radical d is $1/\phi(d)$. Grouping by radical gives $J(R) \geq \sum_{n \leq R} 1/n \geq \log(R+1)$. $\square$

Let $G \rightarrow \infty$, $L = \kappa \log G + O(\sqrt{\log G})$ for fixed $\kappa > 0$, and $S = \lfloor (6/5)LG \rfloor$. The rooted sieve independently excludes one uniform nonzero class modulo each prime $p \leq y$, where $V(y)^{-1}/G \rightarrow 1$ uniformly in the calibrated mixtures. By (7), $y > S$ eventually. Write $\mathcal{U}_z$ for the survivors through z and $0 = x_0 < x_1 < \dots$ for the final survivors.

The global presieve count. Put $z_0 = (\log G)^4$. For $\mathcal{A} = \mathcal{U}_S \cap [1, S]$ and $M = |\mathcal{A}|$,

$$M = (1 + o(1))SV(S)$$

in probability. Fixing the forbidden classes through z_0 , two adjacent Bonferroni truncations of order $b \asymp \log G/(8 \log z_0)$ give $|\mathcal{U}_{z_0} \cap [1, S]| = (1 + o(1))SV(z_0)$ uniformly: CRT rounding costs $O((b+2)z_0^{b+1}) \ll G^{1/3}$, and the reciprocal-product error is at most $2(eT_0/b)^b$, where $T_0 = \sum_{p \leq z_0} p^{-1} = O(\log \log z_0)$. These are $o(SV(z_0))$ and $o(V(z_0))$, respectively. For a candidate $1 \leq a \leq S$, requiring the origin to survive primes above z_0 changes the survival product by a relative $o(1)$, since

$$\sum_{\substack{p > z_0 \\ p|a}} \frac{1}{p} \leq \frac{\log S}{z_0 \log z_0}, \quad \prod_{z_0 < p \leq S} (1 - 1/(p-1)) = \frac{V(S)}{V(z_0)}(1 + O(z_0^{-1})).$$

Thus the conditional mean is $(1 + o(1))SV(S)$, uniformly in the exposed classes. The remaining random coordinates are $z_0 < p \leq S$; their sensitivities $2(S/p+1)$ give conditional variance $O(S^2/z_0 + S \log S + S) = O(S^2/z_0)$. For each fixed $\varepsilon > 0$, Chebyshev therefore gives, eventually,

$$\mathbb{P}(|M - SV(S)| > \varepsilon SV(S)) \ll \frac{1}{\varepsilon^2 (\log G)^2} = o(1).$$

The final subset and its count. Let $U = \mathcal{U}_y \cap [1, S]$ and $N = |U|$. Above S each prime hits at most one candidate, with the same probability for every candidate. Consequently

$$\mathbb{P}(U = E \mid \mathcal{A}, N = n) = \binom{M}{n}^{-1} \quad (E \subset \mathcal{A}, |E| = n) \quad (8)$$

whenever the conditioning event has positive probability. Put

$$\vartheta = \prod_{S < p \leq y} (1 - 1/(p-1)), \quad \vartheta V(S) = (1 + O(S^{-1}))V(y). \quad (9)$$

Exact inclusion probabilities give

$$\begin{aligned} \mathbb{E}(N \mid \mathcal{A}) &= M\vartheta, & \text{Var}(N \mid \mathcal{A}) &\leq M\vartheta, \\ \mathbb{E}\left(\binom{N}{j} \mid \mathcal{A}\right) &= \binom{M}{j} \prod_{S < p \leq y} (1 - j/(p-1)) \leq (M\vartheta)^j/j! \quad (0 \leq j \leq M). \end{aligned} \quad (10)$$

For $j > M$ the moment is zero. Use $1 - jq \leq (1 - q)^j$ with nonnegative factors ($M \leq S < p$); the $j = 2$ case gives nonpositive covariances. The global count and calibration imply $M\vartheta/((6/5)L) \rightarrow 1$ in probability. Conditional Chebyshev gives

$$\mathbb{P}(N < L) = o(1), \quad \mathbb{P}(N > 2M\vartheta) = o(1). \quad (11)$$

On $M\vartheta \asymp L$, the conditional errors are $O(1/L)$. Also $\vartheta G / \log G \rightarrow e^\gamma < 2$.

The auxiliary law $\mathbb{E}^-$ introduced in Section 3.2 has total mass $1 - o(1)$. Uniformly random deletion realizes this law on $N \geq L$: each $(n - 1)$ -subset has $M - n + 1$ extensions. Fixed-rank deletion has a different law.

Lemma 4.2 (Root gaps and auxiliary frames). *At fixed cutoff y , every rooted gap has mean $1/V(y)$. Every fixed collection of r distinct gap ranks below $L - 1$ in U^- , scaled by G , has tight laws and all subsequential limits are dominated by 8^r times Lebesgue measure. The ranks may vary with G ; the assertions are uniform in calibrated mixtures.*

Proof. For $P = \prod_{p \leq y} p$, CRT identifies the rooted environment with a uniform unit $a \bmod P$: its survivors satisfy $(a + x, P) = 1$. The successor map on cyclically ordered units is a permutation, and their cyclic gaps sum to P . Hence, for every rank i ,

$$\mathbb{E}_{0,y}(x_{i+1} - x_i) = P/\phi(P) = 1/V(y).$$

After deleting one point, a gap of rank $i < N - 1$ is at most the sum of the original gaps of ranks $i, i + 1$. This proves tightness for both laws.

Fix r scaled gap intervals $I_1, \dots, I_r$ of positive lengths. For a uniform m -subset, where $m = n - 1$, delete the right endpoints of those gaps. Reinsert them in increasing rank order; each left neighbour is then already known, even for adjacent deleted ranks. If K_i bounds candidate counts in every translate of GI_i , each remaining $(m - r)$ -subset has at most $\prod_i K_i$ preimages. Overcounting by all $\binom{M}{m-r}$ remaining subsets bounds the rectangle probability by

$$\frac{\binom{M}{m-r}}{\binom{M}{m}} \prod_i K_i = \frac{(m)_r}{\prod_{i=1}^r (M - m + i)} \prod_i K_i.$$

Here $(a)_r$ is a falling factorial. On $n \leq M/2$ the bound is $(2/M)^r (n)_r \prod_i K_i$. Now use $\mathbb{E}((N)_r \mid \mathcal{A}) \leq M^r \vartheta^r$ and, for $M > 0$, $\mathbb{P}(N > M/2 \mid \mathcal{A}) \leq 2\vartheta = o(1)$. The case $M = 0$ contributes no mass on $N \geq L$.

The deterministic Selberg upper bound gives $K_i \leq (2 + o(1))|I_i|G/\log G$, uniformly in the interval location and every presieve. Therefore the limiting rectangle mass is at most $(4e^\gamma)^r \prod_i |I_i| < 8^r \prod_i |I_i|$. Portmanteau on open rectangles and regularity give the claimed domination; the missing mass in (11) vanishes. $\square$

Averaging complete frames. Fix an interior resampling rank $j < L$ and let $n \geq L$. For a complete frame $\mathcal{F}$ of size $n - 1$, let $W_{\mathcal{F}}$ be the open interval between its neighbours at the insertion rank j . Every n -subset has a unique decomposition $\mathcal{F} \cup \{u\}$ at this rank. Hence, for $f \geq 0$,

$$\mathbb{E}(f(U) \mid \mathcal{A}, N = n) = \frac{n}{M - n + 1} \mathbb{E}_{\mathcal{F} \text{ uniform}} \sum_{u \in \mathcal{A} \cap W_{\mathcal{F}}} f(\mathcal{F} \cup \{u\}). \quad (12)$$

The expectation includes all $(n - 1)$ -frames, including empty slots. Their original fixed-rank deletion weights are proportional to the number of insertions; (12) keeps this distinction.

On the event $L \leq N \leq 2M\vartheta$, whose complement has mass $o(1)$, eventually $N \leq M/2$ and $n/(M - n + 1) \leq 4\vartheta$. For any nonnegative bounded equicontinuous test family on a compact normalized slot, Selberg also gives

$$\sum_{u \in \mathcal{A} \cap W_{\mathcal{F}}} f_{\mathcal{F}}((u - a_{\mathcal{F}})/G) \leq \frac{3G}{\log G} \left(\int_0^W f_{\mathcal{F}}(z) dz + o(1) \right), \quad (13)$$

uniformly in the retained frames. Here $a_{\mathcal{F}}$ is the left endpoint and $W = |W_{\mathcal{F}}|/G$. To prove this, majorize the test by its suprema on intervals of a fixed normalized length δ . Each has at most $(2 + o(1))\delta G / \log G$ candidates by Selberg. The two cut endpoint intervals cost $O(\delta)$ after normalization. Let G grow, then $\delta \downarrow 0$. With (12), the prefactor is at most $12\vartheta G / \log G < 24$: no division by the interval length is needed.

Lemma 4.3 (Factorial moments). *For $N = |\mathcal{U}_y \cap [1, S]|$, eventually and uniformly in calibrated mixtures and every integer $j \geq 0$,*

$$Q_j := \mathbb{E} \binom{N}{j} \leq (5L)^j / j!.$$

Proof. Lemma 4.1 gives $M = |\mathcal{U}_S \cap [1, S]| \leq (2 + o(1))S / \log S$ in every environment. By (9), calibration and (7),

$$M\vartheta \leq (2e^\gamma + o(1))S/G < 4S/G \leq (24/5)L < 5L.$$

Apply (10) and average; if $j > M$ the conditional moment is zero. No exceptional environment is removed. $\square$

5 From tuple counts to local patterns

5.1 The finite comparison

Let μ, ν be nonnegative masses on subsets of a finite ordered set Ω , with totals m_μ, m_ν . Write $\mu_L(K)$ for the mass whose first L elements are K , b_μ for the mass with fewer than L points, and similarly for ν . Put $p_H = \mu\{U : H \subset U\}$, $q_H = \nu\{U : H \subset U\}$. For $1 \leq L \leq r$ with $r - L$ odd, define

$$\mathfrak{W} = \sum_{j=L}^r \binom{j-1}{L-1} \sum_{|H|=j} \left[(-1)^{j-L+1} (p_H - q_H) \right]_+, \quad Q_j = \sum_{|H|=j} q_H.$$

Lemma 5.1 (First-point comparison for unequal masses). *There is $0 \leq R_\nu \leq \binom{r}{L-1} Q_{r+1}$ such that*

$$\sum_{|K|=L} |\mu_L(K) - \nu_L(K)| + b_\mu \leq b_\nu + 2R_\nu + 2\mathfrak{W} + |m_\mu - m_\nu|. \quad (14)$$

Only the model mass ν supplies the last moment.

Proof. For $|K| = L$ let $B_K = \{v < \max K : v \notin K\}$ and

$$J_\mu(K) = \sum_{\substack{D \subset B_K \\ |D| \leq r-L}} (-1)^{|D|} p_{K \cup D}.$$

On a configuration containing K and m extra earlier points, the alternating sum is 1 if $m = 0$ and $-\binom{m-1}{r-L} \leq 0$ otherwise. Thus $J_\mu(K) \leq \mu_L(K)$, and likewise for ν . Hence

$$[\nu_L(K) - \mu_L(K)]_+ \leq [J_\nu(K) - J_\mu(K)]_+ + \nu_L(K) - J_\nu(K).$$

Each inclusion H of size j occurs for exactly $\binom{j-1}{L-1}$ choices with $\max K = \max H$; the first terms sum to at most $\mathfrak{W}$. For a model configuration of size N , the remaining terms sum to

$$R_{L,r}(N) = \sum_{t=r+1}^N \binom{t-1}{L-1} \binom{t-L-1}{r-L} \leq \binom{r}{L-1} \binom{N}{r+1}.$$

Indeed its summand is $\frac{r-L+1}{t-L} \binom{r}{L-1} \binom{t-1}{r}$. Integrate to obtain R_ν . Finally the exact mass identity

$$\sum_K |\mu_L(K) - \nu_L(K)| + b_\mu = 2 \sum_K [\nu_L(K) - \mu_L(K)]_+ + b_\nu + m_\mu - m_\nu$$

proves the result, including zero masses. $\square$

The left side controls bounded first- L tests and the mass excluded by the actual span restriction, without conditioning on an individual pattern. With $C_* = 5$ and $H_* = C_* + 1 + \log C_*$, Lemma 4.3 gives

$$\sum_{j=L}^r \binom{j-1}{L-1} Q_j \leq e^{C_* L} (C_* L)^L / L! \leq e^{H_* L}. \quad (15)$$

Fix $d_0 = 20$ and take r to be the least integer at least $20L$ with $r - L$ odd. For $L \geq 2$, the exact factorials give

$$R_\nu \leq \frac{(5L)^{r+1}}{(r+1)(L-1)!(r-L+1)!} \leq \frac{1}{20} \left(\frac{(5e)^{20}}{19^{19}} \right)^L = o(1).$$

For the second bound put $q = r - L + 1 \geq 19L + 1$. Using $n! \geq (n/e)^n$ and $(L/(L-1))^{L-1} \leq e$, the first fraction is at most

$$\frac{L}{r+1} (5e)^L \left(\frac{5eL}{q} \right)^q \leq \frac{1}{20} \left(\frac{(5e)^{20}}{19^{19}} \right)^L,$$

since $L/(r+1) \leq 1/20$ and $5e/19 < 1$. The exponential base is below one: $e < 11/4$ and $55^{20} < 4^{20} 19^{19}$. There is no actual moment assumption of order $r+1$.

5.2 The precise one-sided hypothesis

Fix $\kappa > 0$. For primes take $G_X = \log X$, $\rho = e^{1/\kappa}$ and the window parameters (5). Set $N_X = \pi(2X) - \pi(X)$, $\Omega_X = \{1, \dots, S\}$, and let r be the least integer at least $d_0 L$ with $r - L$ odd. For a finite shift set E , the Hardy–Littlewood singular series [6] is

$$\mathfrak{S}(E) = \prod_p \frac{1 - \nu_E(p)/p}{(1 - 1/p)^{|E|}}, \quad \nu_E(p) = |E \bmod p|.$$

For $H \subset \Omega_X$, put

$$C_X(H) = \#\{x \in (X, 2X] : \{x\} \cup (x+H) \subset \mathbb{P}\}, \quad M_X(H) = \mathfrak{S}(\{0\} \cup H) \int_X^{2X} (\log t)^{-|H|-1} dt.$$

Our arithmetic assumption is

$$\mathfrak{E}_X := \frac{1}{N_X} \sum_{j=L}^r \binom{j-1}{L-1} \sum_{\substack{H \subset \Omega_X \\ |H|=j}} \left[(-1)^{j-L+1} (C_X(H) - M_X(H)) \right]_+ \longrightarrow 0 \quad (\text{AHL}_\kappa). \quad (16)$$

The root adds one tuple point. Thus the orders are $L+1, \dots, r+1$, all $O_\kappa(\log \log X)$. At even $j-L$ the adverse error is an undercount; at odd $j-L$ it is an overcount. In particular, the base level $j=L$ requires lower-bound information: this is not a hypothesis of upper sieve bounds alone. The input concerns only prime-tuple counts and their Hardy–Littlewood main terms; it assumes no digital cancellation. A separate error rate for every tuple size is unnecessary. The version on $\{1, \dots, \lfloor 4LG_X \rfloor\}$ implies (16), since its adverse sum contains this one. No optimality among all sufficient hypotheses is asserted.

5.3 Calibration and normalization

The cutoff matches the prime density; weighting cancels the root normalization. An uncalibrated choice such as $y = \sqrt{t}$ would give $V(y) \sim 2e^{-\gamma}/\log t$, off by a constant factor. For each integer $t \in (X, 2X]$, choose the smallest prime $y(t)$ with $V(y(t)) \leq 1/\log t$. Minimality and the Euler-product bound give

$$\frac{1 - 1/y(t)}{\log t} < V(y(t)) \leq \frac{1}{\log t}, \quad y(t) \geq X^\sigma > S$$

for some fixed $\sigma > 0$. Let $Z_X = \sum_t V(y(t))$ and form the probability mixture $\nu_X = Z_X^{-1} \sum_t V(y(t)) \mathbb{P}_{0,y(t)}$. For the comparison keep it unnormalized as $\tilde{\nu}_X = (Z_X/N_X) \nu_X = \zeta_X \nu_X$. Qualitative PNT gives $N_X \sim X/\log X$ and $\zeta_X \rightarrow 1$.

For $|E| = v \leq r+1$ and diameter at most S , distinctness of all residues above $y > S$ gives

$$V_E(y(t)) := \prod_{p \leq y(t)} (1 - \nu_E(p)/p) = \mathfrak{S}(E)(\log t)^{-v} \{1 + O(v^2 X^{-\sigma})\}.$$

Zero local factors give zero on both sides. The integral-to-integer-sum relative error is at most $X^{-1}(\log(2X)/\log X)^v = O(X^{-1})$, by monotonicity. Rooting with weight $V(y(t))$ yields

$$\tilde{q}_H = \frac{M_X(H)}{N_X} (1 + \eta_H), \quad |\eta_H| \leq \delta_X \ll (r+1)^2 X^{-\sigma} + X^{-1}. \quad (17)$$

The actual probability law has $p_H = C_X(H)/N_X$. The positive-part triangle inequality and (15) give

$$\widetilde{\mathfrak{W}} \leq \mathfrak{E}_X + \frac{\delta_X \zeta_X}{1 - \delta_X} e^{H_* L} = o(1).$$

Apply Lemma 5.1, whose bad mass and remainder scale linearly with ζ_X , then normalize the model. Explicitly,

$$\sum_K |\mu_L(K) - \nu_{X,L}(K)| + b_\mu \leq \zeta_X (b_{\nu_X} + 2R_{\nu_X}) + 2\mathfrak{E}_X + 2|\zeta_X - 1| + \frac{2\delta_X \zeta_X}{1 - \delta_X} e^{H_* L} = o(1). \quad (18)$$

This proves (S). The total prime-counting error enters additively. Only the polynomially small calibration error is multiplied by the exponential factor in (15).

A weaker sufficient input. Use the alternating transform J_μ of Lemma 5.1, writing J_{C_X}, J_{M_X} when p_H is replaced by $C_X(H), M_X(H)$, respectively. All sums below have $K \subset \Omega_X$, $|K| = L$. The sufficient input used in Theorem 1.1 is that, for some fixed $c \geq 1$,

$$\mathfrak{D}_{X,c} := 1 - \frac{1}{N_X} \sum_K J_{C_X}(K) + \frac{1}{N_X} \sum_K [J_{C_X}(K) - cJ_{M_X}(K)]_+ \longrightarrow 0. \quad (19)$$

Indeed, for the actual probability law, $A_\mu := 1 - \sum_K J_\mu(K) = b_\mu + \sum_K (\mu_L(K) - J_\mu(K)) \geq 0$. Since $J_\nu \leq \nu_L$, taking positive parts gives

$$b_\mu + \sum_K [\mu_L(K) - c\nu_L(K)]_+ \leq A_\mu + \sum_K [J_\mu(K) - cJ_\nu(K)]_+.$$

The calibration and normalization costs are bounded by c times their previous bounds. Thus (19) yields the positive comparison above, including actual span failure; (T) is unchanged. For $c = 1$, grouping the alternating errors by K before taking positive parts gives $\mathfrak{D}_{X,1} \leq \mathfrak{E}_X + o(1)$: the model remainder and calibration give $N_X^{-1} \sum_K J_{M_X}(K) = 1 + o(1)$. Hence (AHL $_\kappa$) still suffices. Allowing $c > 1$ requires bounded domination, not total-variation approximation. The tuple orders and window are unchanged. No separate $(r+1)$ st factorial-moment assumption is made: actual span failure and the stopped remainder are included in the first term.

5.4 The tail bound and Kuperberg's conjecture

Both arithmetic cases use the same deterministic estimate. If $a_{2n} \leq 4a_n$ eventually, dyadic iteration gives $a_{\lfloor tn \rfloor} \leq 4t^2 a_n$ for $t \geq 1$ and large n . This also ensures polynomial growth and convergence of geometric tails. For $1 \leq u \leq v$, large v and any integer $0 \leq L \leq v$, positivity and telescoping give

$$\begin{aligned} \sum_{n=u}^v T_\rho(n+L) &= \sum_{h \geq 1} \rho^{-h} (a_{v+L+h} - a_{u+L+h-1}) \\ &\leq 4a_v \sum_{h \geq 1} (h+2)^2 \rho^{-h} \ll_\rho a_v. \end{aligned} \tag{20}$$

The bound uses $v+L+h \leq (h+2)v$ and retains the shift $n+L$. For primes, PNT gives $\pi(4X)/\pi(X) \rightarrow 4$ and hence $p_{2n} \leq 4p_n$ eventually. On $I_X = [u, v] \cap \mathbb{Z}$ one has $p_v \leq 2X$, $m_X \asymp X/G_X$ and $L = o(v)$, so (20) proves (T). With $B \geq \rho^D$ equivalent to $D/\log B \leq \kappa$, this proves Theorem 1.1.

Kuperberg's Conjecture 1.3 [7, Conjecture 1.3, equation (7)] posits absolute $K, \epsilon > 0$ such that every admissible distinct $E \subset [0, (\log x)^2]$ with $|E| \leq (\log \log x)^3$ satisfies

$$\left| \sum_{n \leq x} \mathbf{1}_{n+E \subset \mathbb{P}} - \mathfrak{S}(E) \int_2^x (\log t)^{-|E|} dt \right| \leq Kx^{1-\epsilon}. \tag{21}$$

Subtract at $2X$ and X . Every fixed choice of window parameters above is eventually in this range. Inadmissible patterns have both zero main term and zero actual count: a covering prime is at most $|E| < X$. For all others, substituting the raw error directly gives

$$\mathfrak{E}_X \ll GX^{-\epsilon} \sum_{j=L}^r \binom{j-1}{L-1} \binom{S}{j} \leq X^{-\epsilon} \exp(O_\kappa((\log \log X)^2)) \rightarrow 0.$$

No individual main-term lower bound or relative error is needed. Ordinary qualitative Hardy–Littlewood for each fixed tuple does not assert this growing-order, growing-shift uniformity. The singleton case of (21) supplies PNT when assuming Kuperberg's conjecture; the theorem under AHL alone uses the classical unconditional PNT.

6 Scope and formal verification

The present scaling argument balances a local variation amplitude A_{var} at rank $j = \log_B A_{\text{var}} + O(1)$; it needs a growing rank inside the available window and a nonconstant limiting action. A nonzero normal-form class of degree d has amplitude G^d . The identity $p_{n+1}^2 - p_n^2 = 2p_n g_n + g_n^2$ instead retains the absolute position, giving amplitude of order $X \log X$ and rank of order $\log X$, not $\log \log X$; bounded local observables have no growing variation amplitude. This argument does not settle the irrationality of $\sum p_n^2/2^n$. Sequences of positive asymptotic density, such as the squarefree numbers, lie outside the present criterion, whose calibrated scale satisfies $G_X \rightarrow \infty$. The general polynomial argument is qualitative: its compactness and cutoff limits provide no discrepancy rate. For the linear prime series, Appendix B instead gives a finite comparison and an explicit rate under Kuperberg; the qualitative assumption $\mathfrak{D}_{X,c} \rightarrow 0$ alone specifies no rate for its error. Transcendence and irrationality measures are not conclusions.

The dichotomy does not extend to arbitrary integer-valued local functions. The series $\xi_{\text{twin}} = \sum_{n \geq 1} \mathbf{1}_{\{g_n=2\}} 2^{-n}$ has no carries. The classical sieve bound $\#\{p \leq x : p, p+2 \in \mathbb{P}\} \ll x/(\log x)^2$ [24, Theorem 32 and Exercise 34], together with PNT, shows that its ones have density zero; hence it is unconditionally not normal. It is irrational if and only if there are

infinitely many twin primes: an eventually periodic binary sequence of zero one-density is eventually zero, whereas finitely many ones give a rational value. The pair case $E = \{0, 2\}$ of (21) supplies infinitude, and thus an irrational, nonnormal example under Kuperberg's conjecture.

Further directions. Which unbounded integer-valued local observables retain the dichotomy? Examples to examine include logarithmic powers $\lfloor (\log g_n)^\beta \rfloor$ near $\beta = 1$ (before rounding, the local variation has order $(\log G)^{\beta-1}$), rounded piecewise-polynomial expressions such as $\lfloor \min(g_n, t g_{n+1}) \rfloor$ ($t > 0$), and compressed clocks such as $\sum_{n \geq 1} p_{2n} 2^{-n}$: these test the variation scale, flat insertion intervals, and the recurrence's time scale, respectively. A separate question is whether weaker arithmetic assumptions give contiguity of the prime first- L_X pattern law P_X to the calibrated model law Q_X , namely $Q_X(E_X) \rightarrow 0 \Rightarrow P_X(E_X) \rightarrow 0$ for every varying event family E_X , adjoining the same span-failure symbol to both laws. This would rule out concentration on model-negligible families without requiring matching probabilities; these extensions and their tail requirements are left for separate work. Separate follow-up work in preparation develops preliminary normality arguments for a moving-small-prime-cutoff ω -series. These arguments await independent review; normality of $\sum_{n \geq 1} \omega(n) 2^{-n}$ remains unproved here.

Formal verification. A Lean formalization is in progress.

Acknowledgements and provenance. GPT 6 Astra led the mathematical development, and Fable 5.1 acted as a sparring partner.¹

A Unconditional arithmetic for a moving roughness threshold

Let $\mathcal{R}_z = \{n : P^-(n) > z(n)\}$, with any fixed finite initial convention, and write a_n for its increasing enumeration and $g_n = a_{n+1} - a_n$. Set $A(X) = \#(\mathcal{R}_z \cap [1, X])$ and $G_X = V(z(X))^{-1}$.

Proposition A.1 (Local pattern and tail estimates for rough integers). *There is an absolute finite constant A_* with the following property. Fix $\kappa > 0$ and suppose, eventually,*

$$\begin{aligned} z(x) &= \exp(\Psi(\log x)), & \Psi &\in C^1, \quad \Psi(t) \longrightarrow \infty, \\ 0 \leq t\Psi'(t) &\leq C_\Psi \Psi(t), & \frac{t}{\Psi(t)} &\geq A_* \kappa \log(3\Psi(t)). \end{aligned} \tag{22}$$

For any deterministic integer window length $M = \kappa \log G_X + O(\sqrt{\log G_X})$ and $S = \lfloor (6/5)MG_X \rfloor$, short-pattern comparison (S) holds with L replaced by M . Its error is uniform over all bounded tests on these truncated shapes. Moreover

$$A(X) \sim X/G_X, \quad A(2X) - A(X) \sim X/G_X, \quad a_n \ll n \log(n+2), \tag{23}$$

$G_{cX}/G_X \rightarrow 1$ for every fixed $c > 0$, and the actual first-gap condition (T) holds for every fixed auxiliary scale $\rho > 1$.

Proof. We establish the sieve estimate and bound the total error in the pattern comparison.

An explicit growing-dimensional sieve specialization. For a set E of $k \geq 1$ distinct shifts put $\nu_E(p) = |E \bmod p|$ and $V_E(y) = \prod_{p \leq y} (1 - \nu_E(p)/p)$. For a finite interval I of integers, let

¹Fable 5 originally selected Problem 251 in response to the author's question about which Erdős problem it had most enjoyed puzzling over. After initial setbacks, several rounds of encouragement were needed to keep the exploration going.

$C(I, E; y)$ count those $n \in I$ for which every $n + h$, $h \in E$, avoids all primes at most y . There are absolute C_β, C_Q such that, for $y > 4k$, $Z = \lfloor y \rfloor + \frac{1}{2}$, $s = \log R / \log Z \geq 9k + 1$,

$$C(I, E; y) = |I| V_E(y) \{1 + O(e^{-s+C_\beta k})\} + O(e^{C_Q k} R (1 + \log R)^{k-1}). \quad (24)$$

All constants are independent of $k, E, y, |I|, R$. If a local factor vanishes, the count and product are both zero.

Here is the precise import behind (24). Pre-sieve $p \leq 4k$ and put $Q = \prod_{p \leq 4k} p = e^{O(k)}$. There are exactly $Q V_E(4k)$ allowed classes $c \bmod Q$. Set $g(p) = \nu_E(p)/p$ for $4k < p \leq y$, and $g(p) = 0$ otherwise, and extend multiplicatively on squarefree integers. Thus all sums below are supported on divisors of $\prod_{p < Z} p$. Since $g(p) < 1/4$ when it is nonzero,

$$-\log \left(1 - \frac{g(p)}{1 - g(p)}\right) \leq \frac{k}{p} + C \frac{k^2}{p^2}.$$

The reciprocal-prime estimate in (7) and $\sum_{n > 4k} n^{-2} \ll 1/k$ therefore give

$$\prod_{u \leq p < Z} \left(1 - \frac{g(p)}{1 - g(p)}\right)^{-1} \leq e^{Ck} (\log Z / \log u)^k \quad (2 \leq u \leq Z).$$

This verifies the product hypothesis (6-2) of [19, Section 6]. Apply their Lemma 6.2 with $(g', g'') = (g, 0)$ for the lower weights and $(g', g'') = (0, g)$ for the upper weights, with their sieve dimension κ equal to our k . They derive this form from the fundamental lemma of [20, Lemma 6.8]. The beta weights $\lambda_d^\pm$, supported on squarefree $d < R$, have $|\lambda_d^\pm| \leq 1$. They satisfy the sieve inequalities [19, equation (6-4)]. For $\theta^\pm = 1 * \lambda^\pm$ and $h(p) = g(p)/(1 - g(p))$, their explicit bounds are $\sum_d \theta_d^\pm h(d) \leq 1 + e^{9k-s} K^{10}$ for the upper weights and $\geq 1 - e^{9k-s} K^{10}$ for the lower weights, where $K = e^{Ck}$. The exact finite identity

$$\sum_d \lambda_d^\pm g(d) = \prod_{p < Z} (1 - g(p)) \sum_{d | \prod_{p < Z} p} \theta_d^\pm h(d)$$

converts these to the required main-term bounds; no dimension-dependent implicit constant is hidden in this step. Both $p < Z$ and $p \leq Z$ mean precisely $p \leq y$. For squarefree d built from the remaining primes, write $\nu_E(d) = \prod_{p|d} \nu_E(p)$. For each allowed $c \bmod Q$, CRT gives

$$\#\{n \in I : n \equiv c \pmod{Q}, d \mid \prod_{h \in E} (n + h)\} = \frac{|I|}{Q} \frac{\nu_E(d)}{d} + r_{c,d}, \quad |r_{c,d}| \leq \nu_E(d).$$

For squarefree d , $\nu_E(d) \leq d_k(d)$, where d_k counts ordered k -factorizations, and $\sum_{d < R} d_k(d) \leq R(1 + \log R)^{k-1}$ by summing ordered factorizations. The sieve inequalities and summation over the allowed c prove (24), including its entire additive error.

Cutoff variation and density. Write $t = \log X$, $\eta = \Psi(t)$ and $\ell = \log(3\eta)$. By Mertens' asymptotic in (7), $G_X \asymp \eta$, $\log G_X = \ell + O(1)$ and $\ell = O(\log t)$. The range (22) gives $S = O_\kappa(t)$ and $G_X \ll_\kappa t$. Regularity gives $G_{cX}/G_X \rightarrow 1$ uniformly for c in fixed compact subsets of $(0, \infty)$.

For later relative errors we need finer control than this qualitative asymptotic. If $0 \leq \epsilon \log y = o(1)$, then

$$\sum_{y^{1-\epsilon} < p \leq y^{1+\epsilon}} p^{-1} \ll \epsilon + y^{-1/2}. \quad (25)$$

Indeed, for $I = (y^{1-\epsilon}, y^{1+\epsilon}] \cap \mathbb{Z}$, $|I| = O(\epsilon y \log y + 1)$. If $|I| < \sqrt{y}$, count integers. Otherwise the primes in the band all exceed $\lfloor \sqrt{|I|} / \log |I| \rfloor$ eventually. Lemma 4.1 bounds their number

by $O(|I|/\log |I|)$. Since $\log |I| \geq \frac{1}{2} \log y$, dividing by the lower endpoint proves (25). Integer rounding adds only $O(1/y)$.

For a fixed exponent $J > 2$, to be chosen below, partition $(X, 2X]$ into full integer intervals I_i of length $\Delta = \lfloor X t^{-J} \rfloor$ and a discarded final interval of length less than Δ . Put $y_i = z(\min I_i)$ and use actual integer lengths in all sums. Uniformly for $n \in I_i$ and $0 \leq h \leq S$,

$$\log z(n+h) = \log y_i \{1 + O_{C_\Psi}(t^{-J-1})\}.$$

Thus varying-threshold tuple counts lie between the fixed-cutoff counts at $y_i^{1-\epsilon}$ and $y_i^{1+\epsilon}$, where $\epsilon \ll_{C_\Psi} t^{-J-1}$. For every nonzero pattern of size $k = O_\kappa(\ell)$, all primes in this band exceed $2k$ eventually, so (25) gives

$$\frac{V_E(y_i^{1\pm\epsilon})}{V_E(y_i)} = 1 + O\left(kt^{-J-1} + ke^{-\eta/3}\right). \quad (26)$$

A zero local factor occurs at $p \leq k$ and is below both cutoffs eventually; that case remains identically zero throughout the sandwich.

Choose $R = \lfloor X^{1/3} \rfloor$. All sandwich cutoffs are at most $z(X)^2$ eventually, so $s \geq t/(8\eta)$. The $k = 1$ case of (24) now proves

$$N_X := A(2X) - A(X) \sim Z_X := \sum_i |I_i| V(y_i) \sim X/G_X.$$

The discarded interval costs at most $\Delta = o(X/G_X)$ when $J > 2$; the summed endpoint errors are $X^{1/3+o(1)}$. Summing the dyadic estimates down to $X \exp(-\sqrt{\log X})$, with uniformly vanishing relative error, proves $A(X) \sim X/G_X$. In this range regularity and the Euler-product asymptotic give $G_u/G_X = 1 + o(1)$ uniformly; the omitted initial integers are $o(X/G_X)$. Inversion proves (23).

Controlling the total comparison error. Use Lemma 4.3 with $L = M$, where M is the window length of this proposition, not the presieve count of Section 4. Put $C_* = 5$, $H_* = C_* + 1 + \log C_*$ and $d_0 = 20$. Choose r as the least integer at least $d_0 M$ with $r - M$ odd. Every arithmetic pattern in Lemma 5.1 has at most $r + 1$ sites including the root. One possible universal choice is

$$A_* = 8(\max\{9d_0, C_\beta d_0 + H_*\} + 3).$$

Since $M = (\kappa + o(1))\ell$, (22) yields, eventually,

$$s \geq 9(r+1) + 1, \quad -s + C_\beta(r+1) + H_* M \leq -\kappa\ell.$$

Choose the fixed slicing exponent $J > \max\{2, \kappa H_* + 3\}$. The relative tuple error δ_X from the sieve and cutoff sandwich therefore satisfies

$$\delta_X e^{H_* M} \ll e^{-\kappa\ell} + (r+1)(t^{-J-1} + e^{-\eta/3})e^{H_* M} = o(1). \quad (27)$$

The exponential-in- η term is retained even when Ψ grows arbitrarily slowly. The choice of J changes no leading sieve slope.

Including the comparison multiplicities, all tuple sizes, intervals and CRT remainder, the summed additive tuple error is bounded by

$$t^{J+O(1)} 2^{r+1} (1+S)^{r+1} e^{C_Q(r+1)} R(1+\log R)^r = \exp\{t/3 + O_\kappa((\log t)^2) + O_J(\log t)\} = o(X/G_X).$$

To apply the comparison without amplifying a scalar mass discrepancy, let μ_X be the actual configuration mass from the full slices, divided by Z_X , and let

$$\nu_X = Z_X^{-1} \sum_i |I_i| V(y_i) \mathbb{P}_{0,y_i}.$$

Then $m_{\nu_X} = 1$, $m_{\mu_X} = 1 + o(1)$, and for each inclusion $H \subset \{1, \dots, S\}$ the model inclusion mass is

$$Z_X^{-1} \sum_i |I_i| V_{\{0\} \cup H}(y_i).$$

Equation (15), (27) and the additive bound give $\mathfrak{W} = o(1)$ in Lemma 5.1. The model supplies $b_{\nu_X} = o(1)$ and $R_{\nu_X} \leq \binom{r}{M-1} Q_{r+1} = o(1)$. The lemma controls shape discrepancy and actual span failure together. Normalizing μ_X and restoring the discarded interval each costs $o(1)$. This proves (S) for the stated window lengths.

Finally, the density and $G_{4X}/G_X \rightarrow 1$ give $A(4X)/A(X) \rightarrow 4$. At $X = a_n$, this implies $a_{2n} \leq 4a_n$ eventually. For $I_X = [u, v] \cap \mathbb{Z}$ we have $a_v \leq 2X$, $m_X \asymp X/G_X$ and the criterion window $L = O_\rho(\log G_X) = o(v)$. The common estimate (20) proves (T) for every fixed $\rho > 1$. $\square$

For the target degree D and base B , take $\rho = B^{1/D}$ and $\kappa = 1/\log \rho = D/\log B$. Proposition A.1 supplies (S)/(T) with the window parameters (5); Proposition 3.3 and Section 3.3 prove Theorem 1.3.

B Quantitative positive comparison for the prime series

The constant slope of the linear insertion phase permits a finite version of the positive comparison. We keep the same arithmetic quantity $\mathfrak{D}_{X,c}$ from (19), retaining its size instead of only its limit. Write $\text{Lip}(f)$ for the Lipschitz seminorm on $\mathbb{T}$ and $\text{TV}(f)$ for total variation. For $\alpha \in \mathbb{R}$, put

$$D_N^*(\alpha; B) = \sup_{0 \leq t \leq 1} \left| N^{-1} \# \{0 \leq n < N : \{B^n \alpha\} < t\} - t \right|.$$

Lemma B.1 (Finite domination and orbit discrepancy). *Let $B \geq 2$, $m \geq 1$, $a \geq 0$ be integers, $\beta \in \mathbb{R}$, and $\nu = m^{-1} \sum_{n=a}^{a+m-1} \delta_{B^n \beta \bmod 1}$. Suppose that for every nonnegative Lipschitz f on $\mathbb{T}$,*

$$\nu(f) \leq A\lambda(f) + \varepsilon \|f\|_\infty + \eta \text{Lip}(f), \quad A \geq 1, \quad \varepsilon, \eta \geq 0. \quad (28)$$

For every integer $T \geq 1$ and $0 < \delta < 1/4$,

$$\sup_{0 \leq t \leq 1} |\nu([0, t)) - t| \ll_B AT^{-1/2} + \varepsilon + \eta \delta^{-1} B^T + \delta + T/m.$$

Proof. An interval indicator has upper and lower continuous piecewise-linear approximations $\phi^\pm$ on the circle, valued in $[0, 1]$, with variation at most 2, Lipschitz seminorm at most δ^{-1} , and integral errors at most 2δ . For either approximation put

$$F_T(x) = \left| T^{-1} \sum_{h=0}^{T-1} \phi(B^h x) - \lambda(\phi) \right|.$$

Then $\|F_T\|_\infty \leq 1$ and $\text{Lip}(F_T) \leq \delta^{-1} B^T$. The transfer operator $\mathcal{L}_q \phi(x) = q^{-1} \sum_{j=0}^{q-1} \phi((x+j)/q)$ obeys $\|\mathcal{L}_q \phi - \lambda(\phi)\|_\infty \leq \text{TV}(\phi)/q$: compare each summand with the integral on its interval of length $1/q$. Hence

$$\left| \int (\phi(x) - \lambda(\phi))(\phi(B^h x) - \lambda(\phi)) d\lambda(x) \right| \leq 2B^{-h} \quad (h \geq 1).$$

Expanding the square gives $\lambda(F_T^2) \ll_B T^{-1}$ and therefore $\lambda(F_T) \ll_B T^{-1/2}$. Shifting the empirical block by h changes a bounded average by at most $2h/m$, so $|\nu(\phi) - \lambda(\phi)| \leq \nu(F_T) + 2T/m$. Apply (28) to F_T and use the two approximations. The bounds are uniform in the interval. $\square$

Lemma B.2 (The same positive-comparison input at finite scale). *Fix $B \geq 2$, $c \geq 1$ and $\kappa \geq 1/\log B$. Use the prime profiles of Sections 3.1 and 5, with $\rho = e^{1/\kappa}$, $G = \log X$, and the same L, S, r in $\mathfrak{D}_{X,c}$. Put*

$$U(n) = \sum_{i \geq 1} B^{-i} g_{n+i-1}, \quad \mu_X = m_X^{-1} \sum_{n \in I_X} \delta_{U(n) \bmod 1}, \quad \eta_X = G^{-1/2} + GB^{-L}.$$

For every nonnegative Lipschitz f and all sufficiently large X ,

$$\mu_X(f) \leq C_B c \lambda(f) + O_{B,\kappa,c} \left((\mathfrak{D}_{X,c} + L^{-1}) \|f\|_\infty + \eta_X \text{Lip}(f) \right). \quad (29)$$

No convergence or rate assumption on $\mathfrak{D}_{X,c}$ is required for this finite bound.

Proof. First keep the calibrated model unnormalized, $\tilde{\nu}_X = \zeta_X \nu_X$, as in (17). The stopped positive-part inequality giving (19) and the weighted calibration estimate give

$$b_\mu + \sum_K [\mu_L(K) - c\tilde{\nu}_{X,L}(K)]_+ \leq \mathfrak{D}_{X,c} + O_c(\delta_X \zeta_X e^{H^*L}).$$

Indeed, replacing the alternating transform of M_X/N_X by that of $\tilde{\nu}_X$ costs at most the weighted sum of its absolute calibration errors. Here $\zeta_X \rightarrow 1$ by PNT and $\delta_X \ll L^2 X^{-\sigma} + X^{-1}$; for fixed κ the displayed error is $o_\kappa(L^{-1})$. We absorb the bounded factor ζ_X into the positive majorant, rather than paying $|\zeta_X - 1|$ as an error. Thus only an upper model bound remains, with actual span failure included.

The global presieve count has exceptional probability $O_\kappa((\log G)^{-2})$ at any fixed relative tolerance. On its complement, $1.1L \leq M\vartheta \leq 1.3L$ eventually. The conditional variance in (10) therefore gives, uniformly in the mixture,

$$\mathbb{P}(N < L) + \mathbb{P}(N > 2M\vartheta) \ll_\kappa L^{-1}. \quad (30)$$

The deterministic $o(1)$ estimates locating the presieve mean are used only to obtain this fixed tolerance, not as unproved error rates.

Write $\Phi_L(n) = \sum_{i=1}^L B^{-i} g_{n+i-1}$. The deterministic bound (20), with $\rho = B$, gives

$$\text{Avg}_{n \in I_X} |U(n) - \Phi_L(n)| = B^{-L} \text{Avg}_{n \in I_X} U(n+L) \ll_B GB^{-L}.$$

Choose j with $\theta = GB^{-j-1} \in [1, B)$; the profile gives $1 \leq j < L$ eventually. For a complete frame of insertion span GW and left endpoint a , the linear phase is exactly

$$\Phi_L(\mathcal{F}, u) = O_{\mathcal{F}} + bz \pmod{1}, \quad z = (u - a)/G, \quad b = (B - 1)\theta \in [B - 1, B(B - 1)).$$

Partition the normalized slot into intervals of length $\Delta = G^{-1/2}$, with a possible shorter last interval. Lemma 4.1, applied at physical length $\sqrt{G}$, bounds the presieve candidates in each interval by $C\Delta G / \log G$; enlarge the last interval if necessary. Upper sums give

$$\sum_{u \in \mathcal{A} \cap W_{\mathcal{F}}} f(O_{\mathcal{F}} + b(u - a)/G) \leq \frac{C_B G}{\log G} \left(\int_0^W f(O_{\mathcal{F}} + bz) dz + \Delta W \text{Lip}(f) + \Delta \|f\|_{\infty} \right). \quad (31)$$

This estimate is uniform in W . Since $b \geq 1$ and $f \geq 0$ is periodic,

$$\int_0^W f(O + bz) dz \leq (W + b^{-1})\lambda(f) \leq (W + 1)\lambda(f).$$

On $L \leq N \leq 2M\vartheta$, the prefactor in (12) is at most 4ϑ , and $\vartheta G / \log G = O(1)$ uniformly. The auxiliary frame law has mass at most one and $\mathbb{E}^- W \leq 2 \sup_y (V(y)G)^{-1} = O(1)$: realize the law by uniform deletion and bound its rank- j gap by the sum of the two corresponding original gaps, whose rooted means are exact. Charge the exceptional probability (30) before changing measure, average (31), and extend the positive integral to all auxiliary frames. This yields

$$\mathbb{E}[\mathbf{1}_{\{N \geq L\}} f(\Phi_L)] \leq C_B \lambda(f) + O_{B,\kappa} \left(L^{-1} \|f\|_{\infty} + G^{-1/2} (\text{Lip}(f) + \|f\|_{\infty}) \right).$$

The actual positive comparison and tail bound prove (29). $\square$

Corollary B.3 (Quantitative normality under Kuperberg). *Assume (21). For every fixed integer $B \geq 2$, $\alpha_B = \sum_{n \geq 1} p_n B^{-n}$ satisfies*

$$D_N^*(\alpha_B; B) \ll_B (\log \log N)^{-1/2}. \quad (32)$$

Consequently every base- B word of length r occurs among the first N starting positions with count $NB^{-r} + O_B(N/\sqrt{\log \log N})$, uniformly in the word and r . The starting threshold may depend on the constants in (21).

Proof. Choose $\kappa = 4/\log B$. Then $GB^{-L} \leq G^{-3}$. The same odd stopped transform, with $c = 1$, gives

$$\mathfrak{D}_{X,1} \leq \mathfrak{E}_X + O(|\zeta_X - 1| + b_{\nu_X} + R_{\nu_X} + \delta_X \zeta_X e^{H_* L}).$$

This follows also by writing $\mathfrak{D}_{X,1} = 1 - N_X^{-1} \sum_K J_{M_X}(K) + N_X^{-1} \sum_K [J_{M_X}(K) - J_{C_X}(K)]_+$. Under (21), $\mathfrak{E}_X \leq X^{-\epsilon} \exp(O_B((\log G)^2))$. Its singleton case and minimal-cutoff calibration give $|\zeta_X - 1| \ll GX^{-\epsilon} + X^{-\sigma} + G/X$. The model remainder is $O(e^{-c_0 L})$ for an absolute $c_0 > 0$; the calibration term is $o_B(L^{-1})$ and (30) bounds b_{ν_X} . Thus $\mathfrak{D}_{X,1} \ll_B L^{-1}$.

For $q = B - 1$ and $\nu_X = m_X^{-1} \sum_{n \in I_X} \delta_{B^{n-1} \alpha_B \bmod 1}$, Abel and the integer recurrence give $(T_q)_* \nu_X = \mu_X$. For $f \geq 0$, the periodic function $H_f(x) = \sum_{r=0}^{q-1} f((x+r)/q)$ satisfies $f(y) \leq H_f(qy)$, $\lambda(H_f) = q\lambda(f)$, $\|H_f\|_\infty \leq q\|f\|_\infty$, and $\text{Lip}(H_f) \leq \text{Lip}(f)$. Therefore (29) holds for ν_X too, with constants depending on B . No residue equidistribution is needed for this lift. Apply Lemma B.1 with

$$T = \lfloor \tfrac{1}{4} \log_B G \rfloor, \quad \delta = G^{-1/8}, \quad \varepsilon \ll_B (\log G)^{-1}, \quad \eta \ll_B G^{-1/2}.$$

The Lipschitz cost is $O_B(G^{-1/8})$ and $m_X \asymp X/G$, so each prime-value block has star discrepancy $O_B((\log G)^{-1/2})$.

Decompose the prime values up to p_N by the integer-halving construction in Step 4 of Proposition 3.3, stopping below $2\sqrt{p_N}$. Each retained block has $X \geq \frac{1}{2}\sqrt{p_N}$ and hence the common bound $O_B((\log \log p_N)^{-1/2})$. The omitted initial segment and rounding terms number at most $O(\pi(2\sqrt{p_N}) + \log p_N) = o(N/\sqrt{\log \log N})$. Weighted summation and $p_N \sim N \log N$ prove (32). A digit word specifies an interval, whose count error is at most $2ND_N^*$. $\square$

Quantitative and qualitative inputs. For the same $\mathfrak{D}_{X,c}$ at any fixed $\kappa \geq 1/\log B$, Lemma B.2 and the lift above give the finite interface (28) with $\varepsilon \ll \mathfrak{D}_{X,c} + L^{-1}$ and $\eta \ll \eta_X = G^{-1/2} + GB^{-L}$. Taking $T = \lfloor \frac{1}{4} \log_B(1/\eta_X) \rfloor$ and $\delta = \eta_X^{1/4}$ shows that the prime-value block discrepancy is

$$\ll_{B,\kappa,c} \mathfrak{D}_{X,c} + [\log(1/\eta_X)]^{-1/2}.$$

Here $\log(1/\eta_X) \asymp_{B,\kappa} \log G$ if $\kappa > 1/\log B$, and $\asymp_B \sqrt{\log G}$ at equality. The qualitative assumption $\mathfrak{D}_{X,c} \rightarrow 0$ therefore still suffices for normality, but specifies no fixed convergence rate for its own error. Kuperberg supplies the quantitative input used in the corollary. Nonlinear polynomial insertion images need not have bounded densities; a quantitative polynomial classification is not asserted.

C Connections behind the proof

Geometric reindexing underlies both telescoping and the carry recurrence. Figure 4 connects the proof steps with four mathematical perspectives.

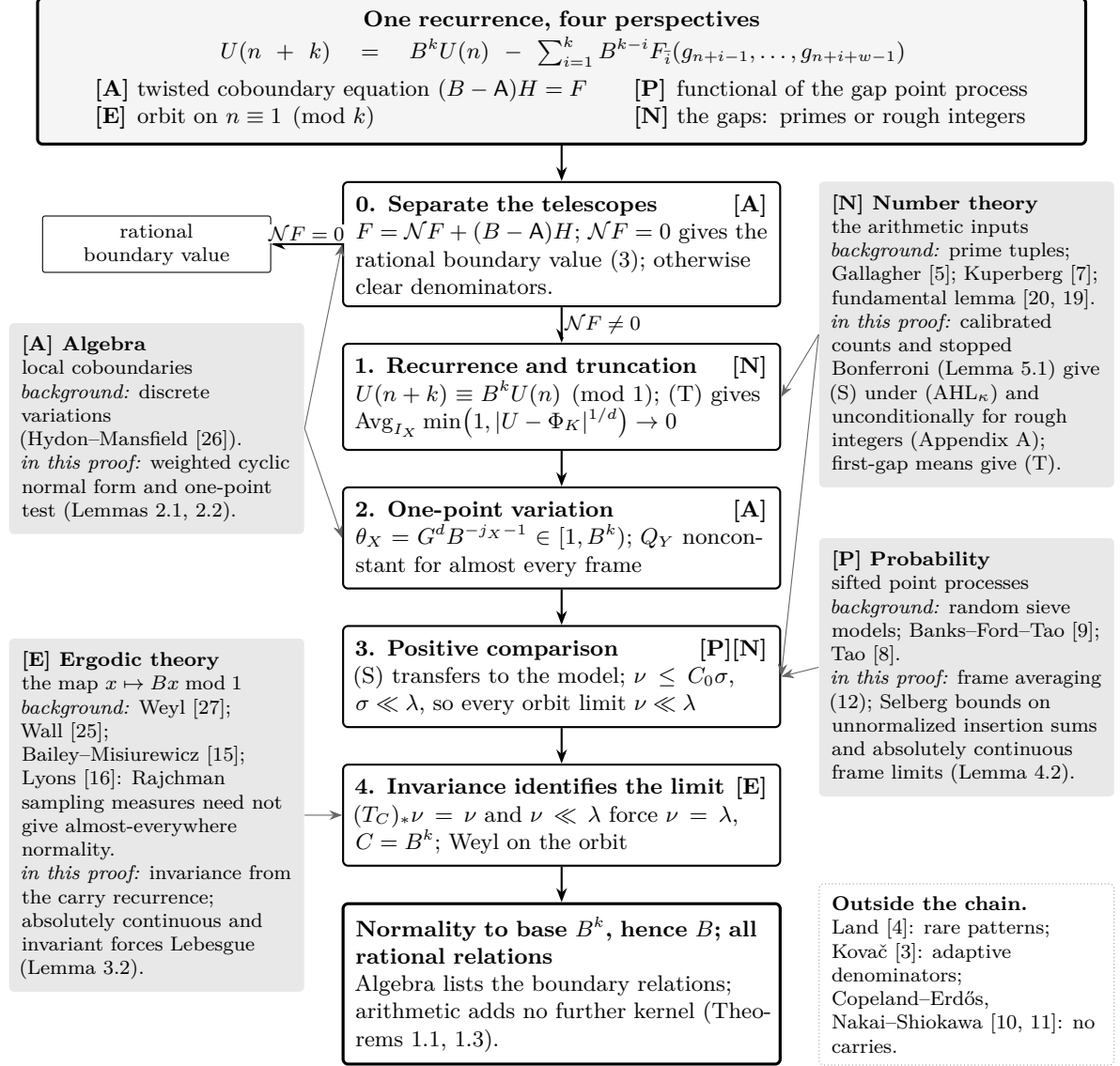


Figure 4: Connections behind the proof. The letters mark the viewpoints used by each step. The spine follows Proposition 3.3 after normal-form reduction, under its hypotheses and degree bound. The shaded blocks distinguish background from its role in this proof; (S), (T) are the two arithmetic inputs. The orbit reading uses integral coefficients and $n \equiv 1 \pmod{k}$, with $U(1) = \alpha_F$. Here ν is an orbit-limit probability measure, λ Lebesgue measure, $T_C(x) = Cx \pmod{1}$, and $C_0 = 24k$. The positive-comparison variant replaces (S) and multiplies C_0 by the fixed factor c .

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